ICT TECHNICIAN L

ICT/OS/IT/CC/01/6

Apply Basic Electronics Skills

WRITTEN ASSESSMENT

TIME: 2 HOURS

INSTRUCTIONS TO CANDIDATE

i. *This paper consists of **TWO** sections: **A** and **B**.*
ii. *Answer **ALL** questions in section A and **ANY THREE** (3) questions in section B.*
iii. *Marks for each question are indicated in the brackets.*
iv. *Candidates are provided with a separate answer booklet*
v. *Do not write on the question paper.*

SECTION A (40 MARKS)

Answer ALL questions in this section

1. A technician is troubleshooting a simple series circuit supplying a light bulb and a resistor Outline a tool that can be used to determine if the circuit is complete. (2 Mark

2. During a lab session, a student measures the current through a resistor in a circuit and finds i lower than expected. Identify FOUR possible causes related to circuit configuration. (4 Mar

3. A diode is connected in a circuit and does not allow current to flow in one direction. U semiconductor theory, identify the phenomenon causing this behavior. (2 M

4. A student is asked to identify components in a simple electric circuit. State FOUR electronic components commonly used in circuits. (4

5. A computer engineer is designing a buffer for data storage. Classify TWO types of suitable for temporary and permanent storage.

6. Convert the decimal number 45 into an 8-bit binary.

7. Illustrate how a NAND gate can be used to implement an AND function using only logic ga
(4 Marks)

8. An LED circuit uses a 9V supply with a 470Ω resistor. Calculate the expected current through the LED. (3 Marks)

9. (a) A student is asked to trace a printed circuit board (PCB) to locate an open circuit. Highlight a two-step systematic approach they would use. (4 Marks)

(b) Outline why a capacitor in a DC circuit initially acts like a short circuit and later blocks current. (2 Marks)

10. Calculate the binary equivalent of hexadecimal 3F. (3 Marks)

11. During a practical test, a student measures voltage drops across resistors in a series circuit. Calculate the expected voltage across a 100Ω resistor in a 12V circuit with 3 resistors: 100Ω, 200Ω, and 300Ω. (4 Marks)

SECTION B (60 MARKS)

Answer THREE questions in this section

12. a) A trainee is required to work with different number systems while developing software for a digital calculator. Explain FIVE disadvantages of using Binary Coded Decimal (BCD) in computers for arithmetic operations. (10 Marks)

b) The rapid development of electronics components has changed the face of the electronic world over the past few years. Explain FIVE emerging trends of such modern electronic components. (10 Marks)

13.

a) Explain THREE ways number systems are applied in computer systems. (6 Marks)

b) Perform the following number systems conversion

156_{10} to:

i. Binary (3 Marks)

ii. Octal (2 Marks)

c) Perform the following number systems arithmetics

i. $11011_2 + 10110_2$ (2 Marks)

ii. $1101_2 - 1011_2$ using 2's Complement (4 Marks)

iii. $101_2 \times 11_2$ (3 Marks)

14. *An ICT technician is servicing several computers with suspected power-related faults.*

a) Explain how voltage, current, and resistance measurements are used together to diagnose electronic problems. (6 Marks)

b) Describe FOUR risks of incorrect use of electronic measuring instruments in ICT maintenance. (8 Marks

c) Propose THREE corrective actions to ensure accurate electronic measurements in future tasks. (6 Mark

15.

a) Outline THREE differences between N-type and P-type semiconductors. (6 Ma

b) Explain the following terms as used in Bipolar transistor configuration. (6 M

i. Common Base Configuration

ii. Common Emitter Configuration

iii. Common Collector Configuration

c) Discuss FOUR types of Read Only Memory. (8

1 A technician can use a multimeter set to continuity mode to check whether the circuit is complete.

2 High resistance in the circuit
Loose or Poor Connections
Incorrect resistor Value.
Low Power Supply Voltage.

3 The behaviour is caused by the PN junction and depletion region. The depletion region allows current in Forward ~~bias~~ bias but blocks it on reverse bias.

4 Resistor
Capacitor
Diode
Transistor

5 RAM: Temporary memory (Volatile)
ROM: Permanent Memory (Non-Volatile)

6 $45 = 32 + 8 + 4 + 1$
$= 00101101_2$

7 A NAND gate can implement an AND function by connecting a NAND gate as a inverter.

Y = [A N AND B] N AND [A

This Produces the same output as an AND gate.

8 Voltage = 9V
Resistance = 470Ω

Using Ohm's Law: $I = \frac{V}{R}$

$I = \frac{9}{4.70}$

$I = 0.0191A$

Ans [19.1mA]

9. Visually inspect the PCB for broken tracks or damaged components. Use a multimeter in continuity mode to trace the circuit and identify the break.

b) A Capacitor initially acts like a short circuit because it is uncharged and allows charging current to flow. When fully charged, it blocks DC current.

10 3 = 0011

F = 1111

= 00111111

11 Supply = 12V

$R_1 = 100\Omega$

$R_2 = 200\Omega$

$R_3 = 300\Omega$

$R_T = 100 + 200 + 300 = 600\Omega$

$I = \frac{12}{600} = 0.02A$

Voltage across 100Ω

$V = IR$

$V = 0.02 \times 100$

$= 2V$

12 It uses more ~~the~~ Memory than binary
It requires more storage bits
Arithmetic operations are slower
It has unused codes from 1010 to 1111
It requires more complex circuits

b) Miniaturization - Components are becoming smaller
Smart sensors - They detect and respond to environmental changes
Low-Power Components - They consume less electricity
Wearable Electronics - Components are used in smart watches and fitness bands
Internet of Things (IoT): Devices can Communic through the internet

13a) Binary - used to process and store Computer data
Hexadecimal - used to represent memory addresses and machine codes
Octal - used in representing binary data and file Permissions

b) 156 = 128 + 16 + 8 + 4
= 10011100_2 Show all the procedure

ii) Divide by 8!
156 ÷ 8 = 19 remainder 4
19 ÷ 8 = 2 remainder 3
2 ÷ 8 = 0 remainder 2
= 234_8

c) $1011_2 + 1011_2$

$$\begin{array}{r} {\scriptstyle 1\,1\,1\,1\,\,\,} \\ 1011 \\ +\,1011 \\ \hline 10110 \end{array}$$

= 10110_2

ii) $1011_2 - 101_2$

$$\begin{array}{r} 1011 \\ -\,0101 \\ \hline 0110 \end{array}$$

2's Complement of 01011

Show all the working

$$\begin{array}{r} 10100 \\ +\ 01011 \\ \hline 11111 \\ +\ 1 \\ \hline 100000 \end{array}$$

$= 10000_2$

iii) $1010_2 \times 11_2$

$$\begin{array}{r} 1010 \\ \times\ 11 \\ \hline 1010 \\ 10100 \\ \hline 11110 \end{array}$$

$= 11110_2$

11a) Voltage - Checks whether the power supply is providing the correct voltage
Current - Helps identify excessive or insufficient current flow
Resistance - Helps detect open circuits, short circuits and faulty components.

b) Electric shock to the technician
Damage to the measuring instrument
Damage to electronic components
Incorrect readings leading to wrong diagnosis

describe!

c) Calibrate measuring instruments regularly
Select the correct meter settings and ranges
Follow safety ~~Procedures~~ Procedures when taking measurements

15 N-type
Uses Pentavalent Impurities
Electrons are majority carriers
Holes are minority carriers

P-type
Uses trivalent impurities
Holes are majority carriers
Electrons are minority carri

b Common Base (CB): The base terminal is shared by the input and output circuits.

Common Emitter: The emitter terminal is shared by the input and output circuits. It provides high amplification.

Common Collector (CC): The collector terminal is shared by the input and output circuits. It is mainly used as a buffer.

c Mask ROM: Programmed during manufacturing and cannot be changed
PROM: Can be Programmed once by the user
EPROM: Can be erased using ultraviolet light and reprogrammed.
EEPROM - Can be erased and reprogrammed electrically